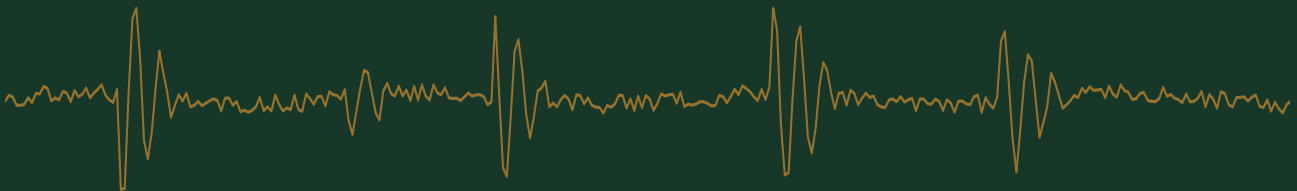




OPEN CATEGORY · THEME 1 — AGRICULTURE

# We hear the threat the tree *can't show*.

An autonomous, solar-powered robotic guardian that detects Red Palm Weevil inside the trunk and treats only the trees that are genuinely sick — no internet, no operator.



RPW FEEDING SIGNAL · 2.4 KHZ

Acoustic signature of Red Palm Weevil larvae inside the trunk

**Team Vcoders**

UNIVERSITY OF PETRA · COLLEGE OF IT · AMMAN, JORDAN

WORLD ROBOT CASPIAN CUP · BAKU

27 JUNE 2026 · PALM-WEB.VERCEL.APP

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THE CORE IDEA

Palm Guard is not just a detector. It closes the robotic loop — **sense → decide → act → alert** — hearing the threat the tree can't show and treating only the trees that are genuinely sick, with no operator and no internet.

|                                     |                                           |                                      |                                          |
|-------------------------------------|-------------------------------------------|--------------------------------------|------------------------------------------|
| <b>3</b><br>SENSORS FUSED ON-DEVICE | <b>0–100</b><br>RISK SCORE, FULLY OFFLINE | <b>~26</b><br>JOD PER PROTOTYPE UNIT | <b>4</b><br>AWARDS & FINALIST PLACEMENTS |
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# Three engineers, one autonomous guardian.

Team **Vcoders** is three undergraduate engineers from the College of Information Technology at the University of Petra in Amman, Jordan, competing in the Open Category under Theme 1 — Technological Solutions for Agriculture. Between us we cover embedded hardware, applied machine learning, full-stack software and commercialisation, which is what let us take Palm Guard from a research idea to a field-mounted prototype that already runs on solar and a Bluetooth mesh, with the full detect-and-treat loop validated on the bench and on real recorded acoustic data.

We split the work by strength but reviewed every major decision together. Our coach, **Dr. Abedal-Kareem Al-Banna** (University of Petra), guided us on process, planning and logistics only; the construction, programming, this report and the booth were built entirely by the team.

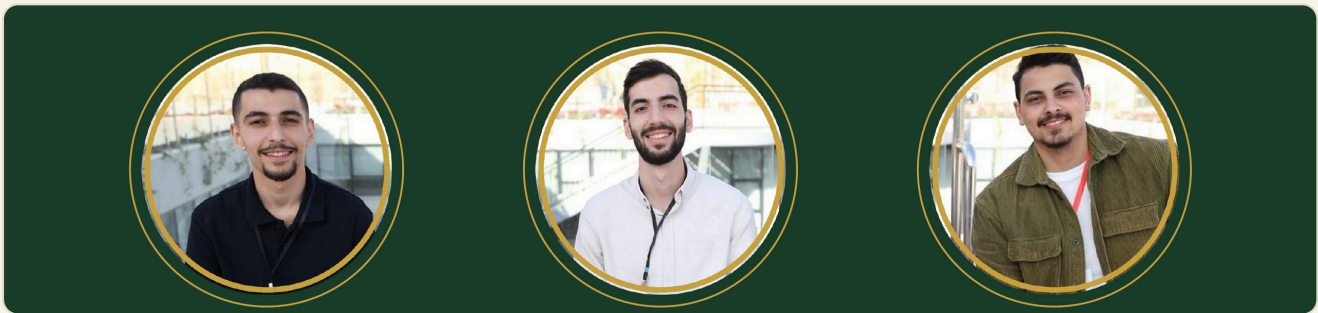
## RECOGNITION

**Best Innovation** — 7th Int'l Jordanian Dates Festival

**1st place** — IEEE Sustainable Technology Hackathon

**2nd place** — AAU 2026 Robotics & AI

**Finalist** — IEEE Region 8 Humanitarian



### Abdelrahman Alkurdi

CTO · TECHNICAL LEAD

Hardware, firmware, on-device AI, enclosure CAD and the injection mechanism.

### Abdelrahman Al-Haimouni

COO · OPERATIONS LEAD

System integration, field testing, Bluetooth-mesh networking, dashboard and backend.

### Zaid Abu Al-Shaar

CBO · BUSINESS LEAD

Problem research, social-impact analysis, business model and presentation.

Figure 1. Team Vcoders — University of Petra, College of Information Technology.

# By the time you can see it, the tree is lost.

The pest, the gap, and what Palm Guard does about it.

The Red Palm Weevil is the most destructive palm pest in the world, and it kills from the inside: the female lays her eggs in the trunk, the larvae tunnel outward unseen, and by the time any external symptom shows the tree is usually beyond saving. It has invaded forty-nine countries and is classified by the FAO as the most dangerous palm pest of all. The damage is done before anyone can see it — exactly the gap Palm Guard closes.

Palm Guard is a low-cost, solar-powered, internet-free robotic device that clamps onto the trunk and runs a complete **sense, decide and act** loop on its own. A steel probe carries three sensors into and against the trunk; the device listens continuously for larval feeding sounds and senses boring vibration and internal temperature rise. An on-device AI ensemble fuses these signals into a single **risk score from 0 to 100**. When an infestation is confirmed, the device does not merely raise an alarm: it activates a peristaltic pump and cannula to deliver a controlled trunk-injection treatment pathway, then reports the event over a Bluetooth mesh to a live dashboard. For the WRCC booth demonstration, the same pump pathway is shown using **clear water only**.

The value is early, targeted, autonomous intervention in place of late visual detection and blanket spraying: Palm Guard catches an infestation weeks before symptoms appear, treats only the trees that need it, and replaces repeated manual scans with one permanent device.

## ALREADY BUILT & BENCH-VALIDATED

Firmware on the ESP32-S3 with a model trained on real recorded acoustic data; a prototype mounted on a palm and running on solar and a Bluetooth mesh; and a complete v4 enclosure and injection system at roughly **26 JOD per unit**. Validating the full loop against a live field infestation is the next step.

Existing tools stop at “there may be a weevil.” Palm Guard detects, decides *and* treats — a permanent guardian that hears what the tree cannot show.

# A clamp-on robot that senses, decides and acts.

A low-cost, solar-powered, internet-free device that runs the whole loop on its own.

Palm Guard clamps onto the trunk and carries three sensors on a steel probe into and against the wood. It listens continuously for larval feeding, and senses boring vibration and internal temperature rise. An on-device AI ensemble fuses these into a single **risk score from 0 to 100**. When an infestation is confirmed it does not merely raise an alarm — it drives a peristaltic pump and cannula to deliver a controlled trunk-injection treatment, then reports over a Bluetooth mesh to a live dashboard. For the WRCC booth, the same pump pathway runs **clear water only**. The value is early, targeted, autonomous intervention in place of late visual detection and blanket spraying — catching an infestation weeks before symptoms appear and treating only the trees that need it.



**Figure 2.** The v4 industrial design, three views — front (RGB status bar and logo), side (service port and the steel sensing probe), and rear (the trunk-injection cannula), with a forward-tilted solar panel. The working prototype is built and bench-validated, running on solar and a Bluetooth mesh.

|                                                                                                  |                                                                      |                                                              |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Built &amp; bench-validated</b><br>FIRMWARE ON ESP32-S3 · MODEL ON A<br>REAL ACOUSTIC DATASET | <b>~26 JOD / unit</b><br>COMPLETE V4 ENCLOSURE + INJECTION<br>SYSTEM | <b>Solar + BLE mesh</b><br>OFF-GRID, INTERNET-FREE OPERATION |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------|

# From a single microphone to a field robot.

How acoustic-only detection became a self-developed robotic system — by deliberate engineering choices.

Palm Guard began as an **acoustic-only** detector: a single microphone listening for larvae feeding inside the trunk. On the bench it worked — but the field broke it. Wind, irrigation, machinery and ambient farm noise masked the faint feeding sounds, and one microphone cannot tell a gust from a grub. Acoustic evidence alone produced too many false positives to justify an automated response.

So we changed the architecture on purpose. Each weakness pointed to another sensor: **vibration** to confirm boring in the wood, and **temperature** to catch metabolic heat. We fused them with an on-device AI ensemble, added a Bluetooth mesh and a dashboard for alerts, and closed the loop with a controlled trunk-injection pathway. What started as a sensor became an **autonomous robotic node** that senses, decides, acts and reports.

## DECISIONS

### What we rejected, and why.

| WE CONSIDERED                       | WE REJECTED IT BECAUSE                                          | WE CHOSE INSTEAD                                        |
|-------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------|
| Cloud-only AI                       | Many farms have weak or no internet                             | On-device edge AI, fully offline                        |
| Manual inspection & trap scouting   | The infestation is hidden inside the trunk until it is too late | Continuous internal multi-sensor sensing                |
| Acoustic-only as the decision layer | One sensor creates false positives in field noise               | Three-signal fusion with a 0-100 confidence score       |
| Fully automatic chemical injection  | Unsafe and uncontrolled without oversight                       | Autonomous loop + an operator safety gate for pesticide |

#### THE TAKEAWAY

The transformation was intentional and engineering-driven. Every rejected idea pushed Palm Guard further from a monitoring gadget and closer to a **self-developed robotic solution** — built, not bought.

# The autonomous loop — decide, then act.

Not a fixed-sequence machine: it acts only when the evidence converges.

|                                                                           |                                                                     |                                                       |                                                                    |                                                                    |
|---------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>01</b><br>● <b>Sense</b><br>Three sensors read the trunk continuously. | <b>02</b><br>● <b>Extract</b><br>MFCC + spectral + sensor features. | <b>03</b><br>● <b>Score</b><br>Ensemble → risk 0-100. | <b>04</b><br>● <b>Agree</b><br>Threshold + multi-sensor agreement. | <b>05</b><br>● <b>Act</b><br>Inject & report — or keep monitoring. |
|---------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|

We started from one hard fact: with this pest, by the time you can see the problem the tree is usually lost, so early detection from inside the trunk is the whole game. Two decisions shaped Palm Guard. First, rather than trusting one signal, we fuse **three independent inside-trunk signals** with on-device AI to suppress false alarms. Second — and this is what makes Palm Guard a robot rather than a sensor — we **close the loop**: the device decides, then acts on its own by injecting treatment, with no operator and no internet.

## HOW PALM GUARD DIFFERS

### Every competitor stops at detection.

| CRITERION                       | PALM GUARD                        | PALMEAR       | IOTREE (AGRINT) | TRAPS / VISUAL |
|---------------------------------|-----------------------------------|---------------|-----------------|----------------|
| Detection method                | Acoustic + vibration + thermal AI | Acoustic + AI | Seismic sensor  | Visual / traps |
| Pre-symptom detection           | ✓                                 | ✓             | ✓               | ✗              |
| Continuous 24/7                 | ✓                                 | ✗             | ✓               | ✗              |
| Autonomous, no manual scan      | ✓                                 | ✗             | ✓               | ✗              |
| Automatic treatment of the tree | ✓                                 | ✗             | ✗               | ✗              |
| Solar / off-grid                | ✓                                 | ✗             | ~               | ✗              |
| Connectivity                    | Bluetooth mesh                    | App per scan  | Cloud gateway   | —              |

One permanent device replaces repeated scans and labour for a lower total cost of protection. To our knowledge, no existing product offers the autonomous treatment stage.



# Two zones and a head — a clean drop-in build.

A self-developed edge device — not an off-the-shelf kit.

**Zone A** faces the palm and carries a steel probe inserted into the trunk, an acoustic sound guide, the microphone, the temperature and vibration sensors.

**Zone B** holds the electronics: the ESP32-S3 controller, the power chain (LiPo, CN3065 solar charger, TPS63802 regulator), the peristaltic pump, reservoir and cannula, and the RGB status light. The **head** carries a forward-tilted solar panel aimed at the sun.

The v4 design is mechanically complete: closed base, real sensor openings, corrected solar orientation, verified for interference and sealing, split into three printable parts (Body, Lid, Probe) with full STLs and a build manual.

Prototype BOM  $\approx$  **26 JOD**.

DESIGN PRINCIPLE

*Engineering care, not just control* — an acoustic waveguide couples the inside of the trunk to the mic so faint feeding sounds reach the sensor cleanly; the probe and the treatment cannula both penetrate the trunk.

TABLE 1 · PROTOTYPE BILL OF MATERIALS BY SUBSYSTEM —  $\approx$  26 JOD AT PROTOTYPE QUANTITIES; TARGET  $\approx$  10 USD / UNIT AT SCALE

|            |                                                                                                                                           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| CONTROLLER | ESP32-S3 — WiFi and Bluetooth LE on-chip                                                                                                  |
| SENSING    | INMP441 acoustic mic · SW-420 vibration · DS18B20 waterproof temperature                                                                  |
| POWER      | LiPo 3.7 V · 5 V solar panel (135×64.7 mm) · CN3065 solar charger · TPS63802 buck-boost (3.3 V) · 220 $\mu$ F capacitor · KCD1-105 switch |
| ACTUATION  | 5 V micro-peristaltic pump · IRF540 MOSFET (33 A / 100 V) · 2×4 mm silicone tube · RGB status LED                                         |
| COMMS      | Bluetooth LE mesh — built into the ESP32-S3, no separate radio module                                                                     |
| BUILD      | Double-sided PCB, 9×15 cm                                                                                                                 |



# Inside the device — and the stack behind it.

A clean drop-in enclosure, and a self-developed software / hardware stack.

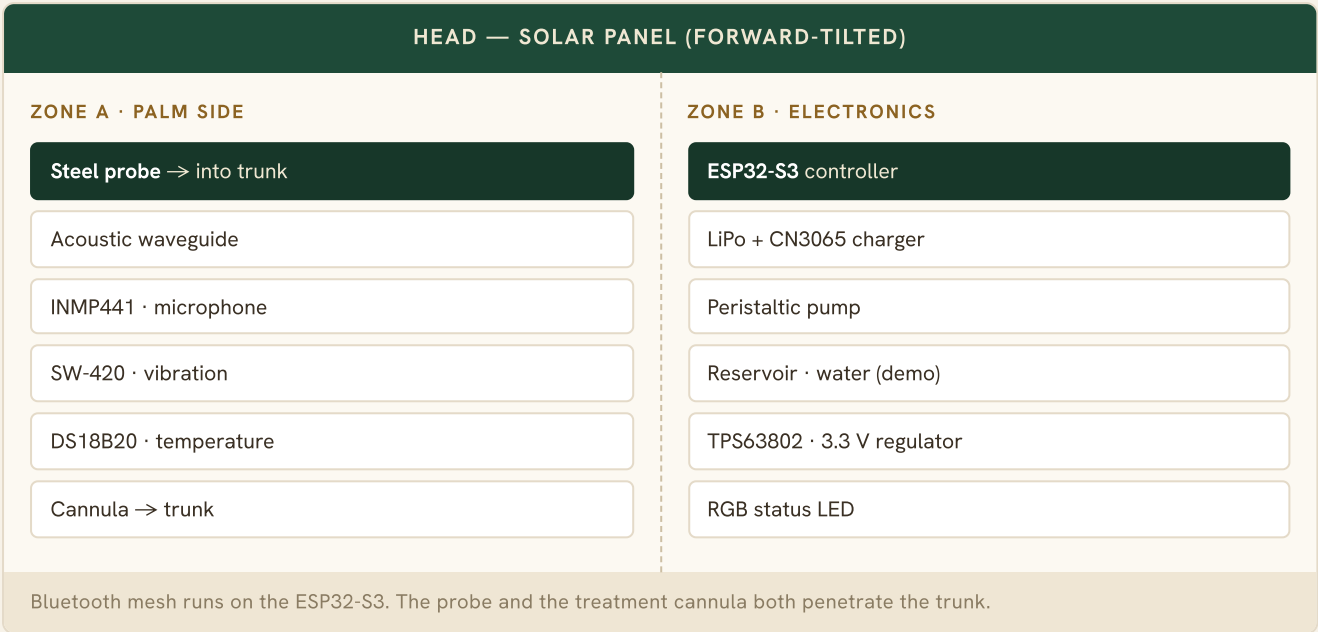


Figure 3. Enclosure section — Zone A couples to the trunk through a steel probe and an acoustic waveguide and carries the three sensors; Zone B holds the controller, power, pump, reservoir, regulation and the status LED.

## SYSTEM ARCHITECTURE

# A custom multi-model pipeline, not a kit.

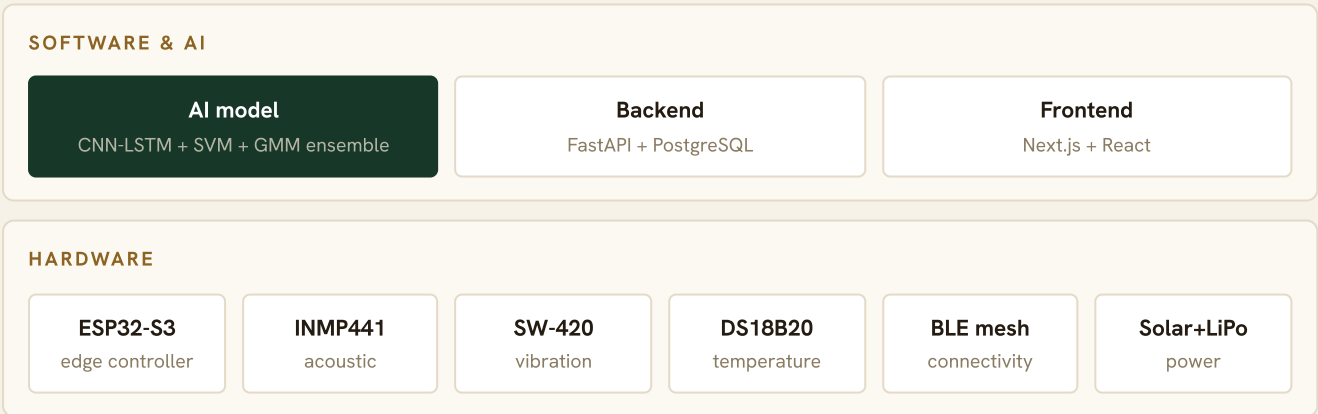


Figure 4. System architecture — a self-developed edge device with a custom multi-model pipeline, not an off-the-shelf kit.

# Three senses inside the trunk, fused on-device.

### INMP441 · MICROPHONE

Low-amplitude larval feeding and boring sounds inside the trunk — energy in the low band with peaks near 2.4 kHz.

### SW-420 · VIBRATION

The mechanical vibration produced as larvae bore through the wood.

### DS18B20 · TEMPERATURE

Internal trunk-temperature rise from larval metabolic activity.

### WHY THREE, NOT ONE

A single microphone confuses wind with feeding. Requiring agreement across three independent signals lets the model confirm an infestation only when the evidence converges — which sharply reduces false positives, the main weakness of single-microphone detectors.

## WIRING OVERVIEW

# Everything on one controller.

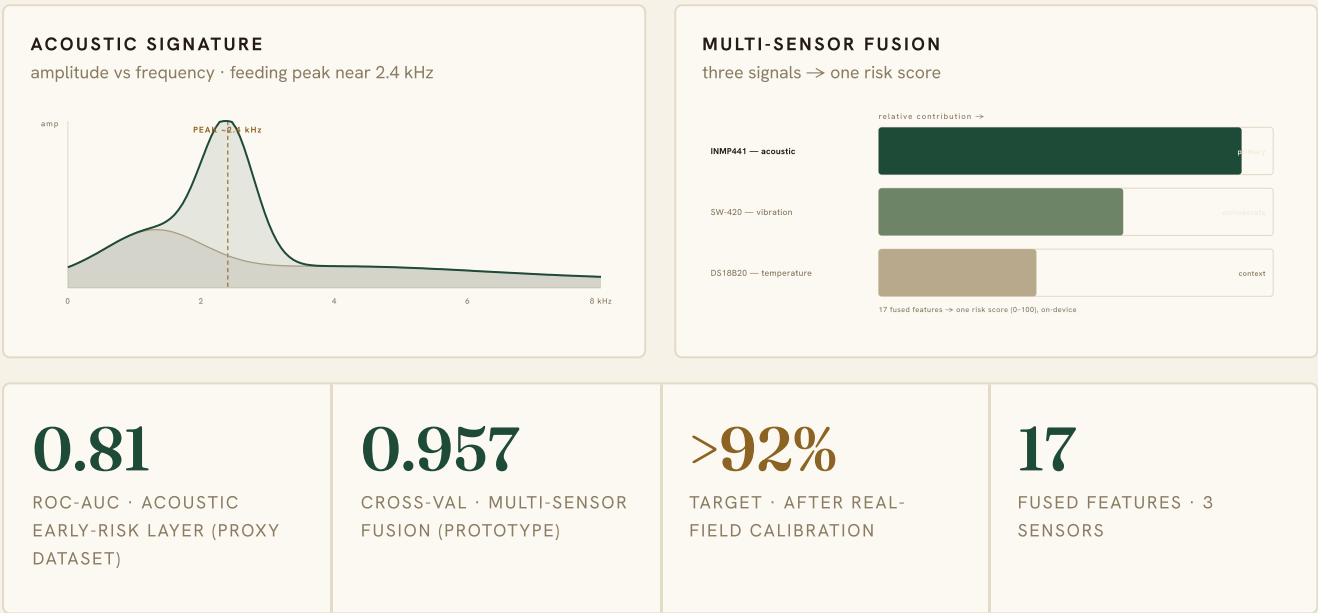


Figure 5. Wiring overview — every sensor and actuator connects to the ESP32-S3 over a standard bus; the treatment pump is driven from GPIO38 through an IRF540 MOSFET.

# A hybrid ensemble, running at the edge.

Inference on the ESP32-S3 — offline, low-latency, low-power and private.

Audio is framed and converted to MFCC and spectral features; dynamic thresholding and noise reduction clean the signal; and a hybrid ensemble combines three complementary models — a **Gaussian Mixture Model** for acoustic background and anomaly, a **Support Vector Machine** for feature classification, and a **CNN-LSTM** for the spectro-temporal pattern of feeding bursts. The acoustic result is fused with the vibration and temperature features — seventeen in total — into one risk score.



*Palm Guard does not treat the 0.81 ROC-AUC acoustic layer as a final diagnosis: it uses acoustic analysis as an **early-risk signal** inside the fusion pipeline, where confidence rises only when vibration, temperature and sensor-health checks agree. The 0.81 is an initial result on a **proxy-species dataset** (real RPW field recordings are scarce); 0.957 is the fused-ensemble cross-validation; >92% is the **target after real-field calibration**, not a measured field result. Fusing signals turns a limited single-sensor classifier into a stronger, safer robotic decision system.*

# The loop, as compiled firmware.

Listing 1 — PlatformIO / C++ running locally on the ESP32-S3, published open-source by the team.

## LISTING 1 · SENSE → DECIDE → ACT LOOP — ESP32-S3, FIRMWARE V3.0

```
// Palm Guard — sense → decide → act loop (ESP32-S3, firmware v3.0)
void loop() {
  SensorFrame f = readSensors();           // INMP441, SW-420, DS18B20
  float feat[17];
  extractFeatures(f, feat);                 // MFCC + spectral + sensor features
  float risk = ensemble.score(feat);       // GMM + SVM + CNN-LSTM → 0..100
  if (risk >= RISK_THRESHOLD && sensorsAgree(f)) {
    injectTreatment();                     // GPIO38 → IRF540 → pump; demo = clear water
    mesh.send(buildAlert(risk, f));        // report over Bluetooth mesh
    setStatus(ALERT);                      // status LED → red
  } else {
    setStatus(MONITORING);                 // status LED → green
  }
  sleepUntilNextWindow();                  // duty-cycle for solar / battery
}
```

### EMBEDDED ON THE ESP32-S3

The full pipeline — MFCC features, the GMM/SVM/CNN-LSTM ensemble and the risk score — is compiled to run on the microcontroller itself, so inference happens at the edge with no cloud call.

### ACTS WITHOUT A NETWORK

When the score crosses threshold and multiple sensors corroborate, the robotic node drives the treatment pump and reports over Bluetooth mesh — on its own, with no cloud in the path.

### PUBLISHED

The full firmware and the live dashboard are published by the team. Live monitoring: [palm-web.vercel.app](https://palm-web.vercel.app).

## It treats only when the evidence agrees.

The conditional decision-making that makes this a robot, not a repeating machine.

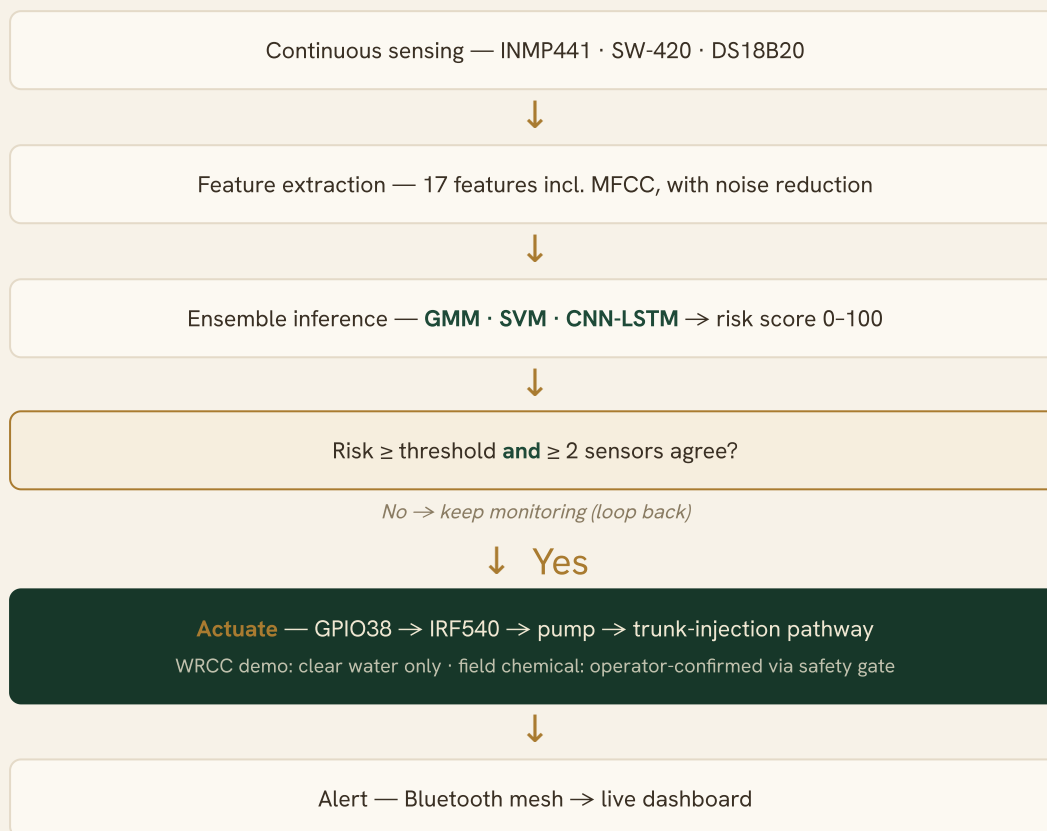


Figure 6. On-device decision logic — the device treats only when the risk score crosses threshold and multiple sensors corroborate, then acts and reports locally, with no cloud or remote control.

### WHY THIS IS A ROBOT, NOT A SENSOR

Every competitor stops at “there may be a weevil.” The conditional branch above — act *only* on agreement, otherwise return to monitoring — is what makes Palm Guard an autonomous robotic node: it senses, decides, acts and reports on its own.

# Closing the loop — the part that sets us apart.

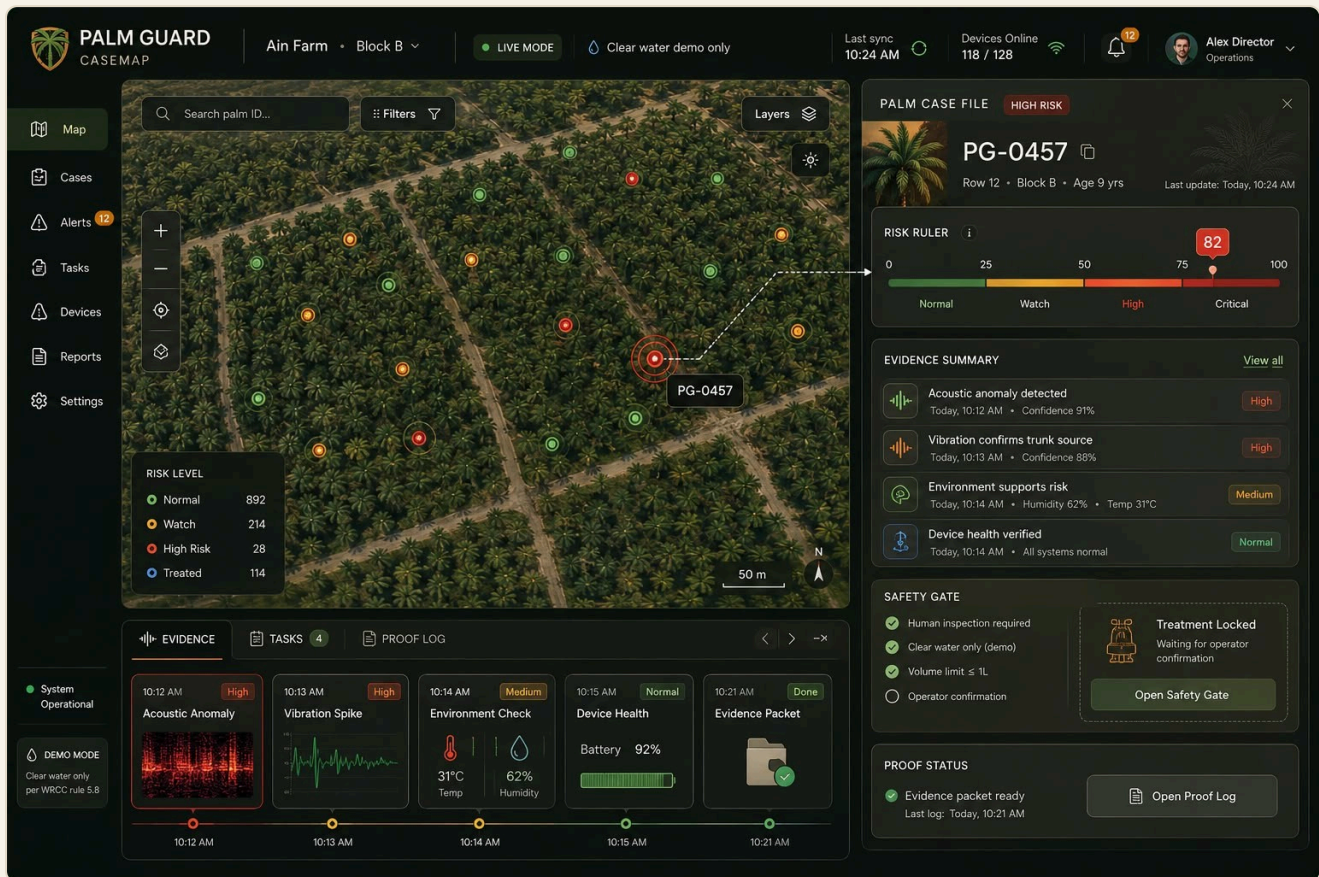
When risk is confirmed, the ESP32-S3 drives a control line through an IRF540 MOSFET to a 5 V peristaltic pump, moving liquid from the onboard reservoir through a cannula into the trunk beside the sensing probe. **Detection and decision run fully autonomously** — no remote control, internet or operator — and the booth demonstrates the complete loop live with clear water. Trunk micro-injection of emamectin benzoate is an **established** preventative and curative method, confirmed in a field trial in Jordan by the National Agricultural Research Centre. For real chemical treatment in the field, an operator confirms the dose through a **safety gate** before any pesticide is injected — a human stays accountable for the regulated step, and every event is logged for traceability. Palm Guard's contribution is delivering a proven treatment pathway **automatically and early**, the moment the model confirms risk.

| FUNCTION    | WRCC BOOTH DEMONSTRATION             | REAL AGRICULTURAL DEPLOYMENT           |
|-------------|--------------------------------------|----------------------------------------|
| Liquid used | Clear water only                     | Approved trunk-injection treatment     |
| Purpose     | Safely show autonomous actuation     | Treat a confirmed infestation          |
| Trigger     | AI risk threshold / demo pattern     | AI risk threshold from field sensors   |
| Human role  | No remote control during actuation   | Operator confirms dose via safety gate |
| Safety      | No pesticide, smoke or unsafe liquid | Follows agricultural regulation        |
| Output      | Pump · LED · mesh alert · dashboard  | Treatment log · alert · history        |

THE LIVE BOOTH DEMONSTRATION

In the 2×2×2 m booth, judges see the complete sense-decide-act-alert loop run safely: the device streams readings, scores risk, updates the dashboard, then autonomously actuates the pump with clear water as the status LED changes and a mesh alert appears. Because treatment is targeted only to confirmed trees as a precise in-trunk dose, Palm Guard reduces the need for broad insecticide spraying.

# Offline-first, solar-powered, always watching.



**Figure 7.** The live monitoring dashboard (palm-web.vercel.app): a per-palm **case file** with a 0–100 risk ruler and a traceable **evidence packet** — acoustic, vibration and environmental signals with confidence scores — plus a **safety gate** that holds the treatment dose for operator confirmation. Demo mode runs clear water only, per WRCC rule 5.8.

## POWER & CONNECTIVITY

The ESP32-S3's built-in Bluetooth LE links units in a low-power mesh and forwards events to a nearby gateway with no cellular network — the system is offline-first.

A forward-tilted 5 V solar panel, LiPo, CN3065 charger and TPS63802 regulator give effectively zero grid draw and continuous round-the-clock operation.

The dashboard presents a farm digital twin, real-time alerts with the signal behind each flag, and a mesh & device-health view across the network.

## CHALLENGES WE SOLVED

- Tiny insect sounds vs ambient noise → multi-sensor fusion, MFCC features and dynamic thresholding
- Running an ensemble on a microcontroller → model compression to fit the ESP32-S3
- Trunk coupling & sealing → resolved in the v4 drop-in assembly
- Off-grid autonomy → solar power and low-power BLE duty cycling
- Safe live demonstration → clear-water booth pathway, separate from real treatment



# Built from real parts, for real farms.

Off-the-shelf where it should be, original where it matters — with the numbers behind it.

Palm Guard is built from proven, off-the-shelf parts — the ESP32-S3, standard sensors, a small solar panel and on-chip Bluetooth — and that is a **deliberate engineering choice, not a shortcut**. These parts are affordable, available, repairable and realistic for a farmer to own and maintain. The originality is not in inventing a microphone; it is in the **integration**: the firmware, the multi-sensor fusion, the sealed enclosure and power chain, the edge-AI decision logic, the mesh alert workflow and the robotic actuation — assembled into one self-developed node that runs off-grid. That is what makes it scalable.

| SENSOR / LAYER     | WHAT IT DETECTS                             | FALSE POSITIVE IT REDUCES               |
|--------------------|---------------------------------------------|-----------------------------------------|
| Acoustic           | Internal feeding sounds & abnormal activity | Reliance on late visual symptoms        |
| Vibration          | Trunk vibration & boring confirmation       | Wind, handling & random acoustic spikes |
| Temperature        | Trunk & environment temperature context     | Heat-related misreads                   |
| Sensor-health gate | Faulty or abnormal readings                 | A damaged sensor raising false alerts   |
| Fusion model       | All evidence → one risk score               | Any single-sensor mistake               |

Table 1. How each layer narrows false positives — the engineering reason Palm Guard fuses three signals instead of trusting one.

POWER BUDGET · ESTIMATED PROTOTYPE VALUES

## Solar-sustained by duty-cycling.

| MODE              | ESTIMATED CURRENT | NOTE                                |
|-------------------|-------------------|-------------------------------------|
| Deep-sleep / idle | < 1 mA avg        | RTC wake between cycles             |
| Active sensing    | ~80-120 mA        | Brief window per cycle              |
| BLE mesh transmit | ~100-130 mA peak  | Milliseconds per report             |
| Daily energy      | ~50-150 mAh / day | Duty-cycled — to be field-validated |

Estimated prototype values, to be confirmed in extended field testing. The forward-tilted 5 V panel, LiPo, CN3065 charger and TPS63802 regulator supply well above this budget in daylight, so the node runs round-the-clock with effectively zero grid draw — duty-cycling keeps the average current low enough for solar to sustain indefinitely.

# A pillar of food security, under attack.

49

COUNTRIES INVADED

€483M

DAMAGE / YR ·  
MEDITERRANEAN

80%

OF ITS LIFE HIDDEN IN  
THE TRUNK

25–40%

HARVEST LOST  
UNTREATED

The FAO classifies the Red Palm Weevil as a category-1 pest and the most dangerous palm pest in the world; it attacks more than forty palm species. Because the larvae feed inside the trunk, an infestation stays invisible until the tree is dying — and a mature date palm is six or more years of growth, worth on the order of a thousand US dollars. For Jordan the threat is immediate: the weevil reached the Jordan Valley around 1999, and at the Azraq oasis (~50,000 palms) about 1,000 infested trees were removed in a single campaign. Critically, the trunk-injection treatment Palm Guard automates was validated in Jordan itself.

Instead of spraying an entire grove, Palm Guard treats only the confirmed, infected tree with a precise micro-dose — *green high-tech* that works with the living system rather than against it.

## RESPONSIBLE DESIGN

**False positives:** needs high risk +  $\geq 2$  sensor groups; optional human-confirm mode.

**Chemical:** sealed reservoir, local regulation, agronomy supervision.

**Data:** offline-first; farm data stays local.

## Food

A DIRECT PILLAR OF ARAB  
FOOD SECURITY

## Economic

DECADES OF PRODUCTION  
LOST PER TREE

## Social

PRIMARY INCOME FOR  
JORDAN VALLEY FARMERS

## Environment

PALMS RESIST  
DESERTIFICATION

## THE KEY INNOVATION

Palm Guard is a low-cost, fully autonomous, internet-free robotic system that both *detects* Red Palm Weevil activity through multi-sensor sensing and an on-device AI ensemble, and *acts* on it through a trunk-injection pathway — demonstrated autonomously with clear water, with an operator safety gate for chemical treatment in the field. Existing solutions stop at detection; Palm Guard closes the loop.

# From signal to a saved tree.

What the loop looks like on a real farm — and the platform behind it.

Consider a date farm in the Jordan Valley. Palm Guard units are clamped to the trunks and form a Bluetooth mesh back to a single gateway, and the farmer watches a live dashboard of every tree.

On one palm, the device picks up a high-frequency vibration and acoustic signature consistent with larval feeding — a peak near 2.4 kHz — together with a temperature rise above baseline. The fused risk score crosses the confirmation threshold, the device triggers a controlled trunk-injection treatment according to the approved field protocol, and an alert appears on the dashboard. The tree is treated **weeks before** any external symptom would have shown — where today’s practice would catch it only when the fronds begin to collapse, too late to save.

04 · PRODUCT ECOSYSTEM

## A connected platform, not a single device.

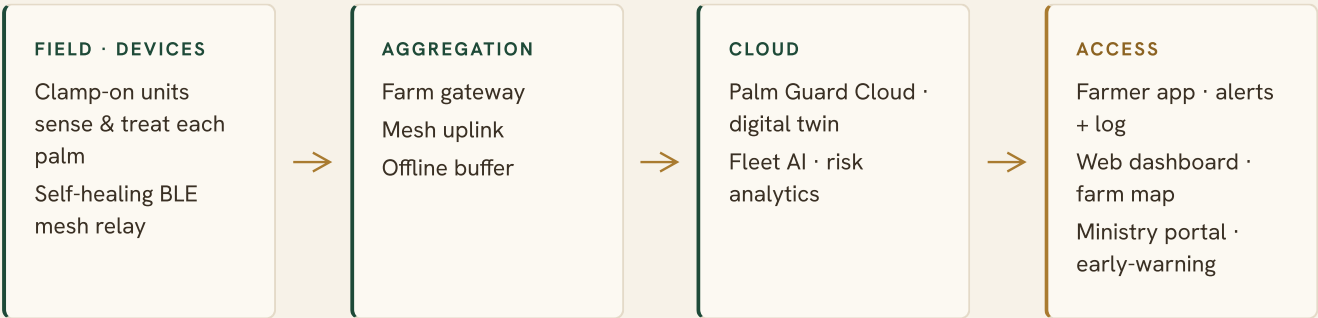
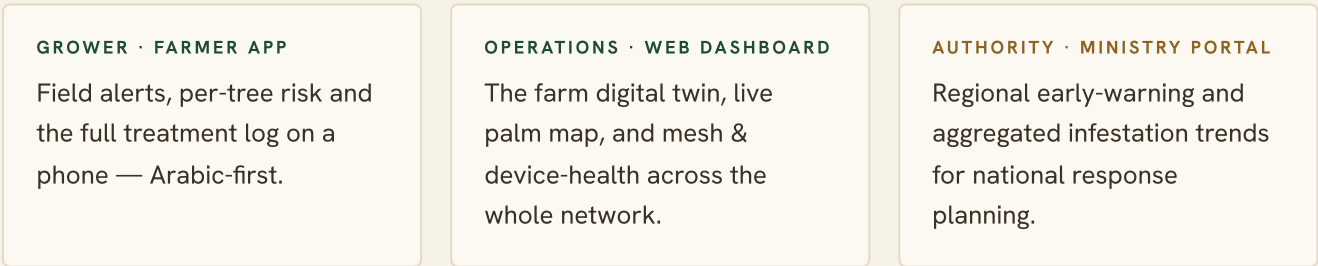


Figure 8. The Palm Guard platform — data flows from autonomous clamp-on units up through a self-healing Bluetooth mesh and farm gateway to the cloud digital twin, then out to the farmer app, the web dashboard, and a ministry early-warning portal.



WHY A PLATFORM, NOT A PRODUCT

One device protects a tree; the mesh protects a region — per-tree detection becomes farm-wide and ultimately national early warning, with farm data kept local and offline-first.

# Built to become a real product.

~10 USD to make · \$20 unit + per-tree SaaS · 200–350 USD saved per tree.

## 1 CUSTOMER SEGMENTS

- Palm farms — small / medium / large
- Municipalities & Ministry of Agriculture
- Landscaping companies & resorts
- Gulf high-value date farms (premium)
- Agricultural contractors & pest-control

## 2 VALUE PROPOSITION

- Early detection **20–30 days** before symptoms
- Prevents **70%+** loss — avoids ~\$500/tree
- Multi-signal AI: acoustic + vibration + thermal
- Fully autonomous — self-decides & auto-delivers treatment
- Solar-powered robotic node — first in Jordan

## 3 CHANNELS

- Direct field sales to farms
- Government pilot → national rollout
- Agricultural exhibitions & unions
- WhatsApp outreach + on-farm demos
- Distributor partnerships (Gulf entry)

## 4 RELATIONSHIPS

- Automated alerts & proactive monitoring
- Dedicated farmer support line (Arabic)
- In-dashboard onboarding agent
- Annual maintenance visits
- Training & awareness workshops

## 5 REVENUE STREAMS

- Hardware: **\$20** unit · ~\$150 gateway/farm
- SaaS / tree / mo: **\$0.80 / \$0.60 / \$0.40**
- Pesticide cartridge refills (recurring)
- Installation, maintenance & data tier
- Small ≤200 · Medium 201–1,000 · Large 1,000+

## 6 KEY RESOURCES

- AI models & sensor-fusion tech
- R&D hardware & embedded systems
- Pesticide injection mechanism
- Palm dataset & field knowledge
- Engineering & field-ops team

## 7 KEY ACTIVITIES

- Continuous AI training & improvement
- Hardware manufacturing & QA
- Field deployment & onboarding
- Auto-detection & auto-treatment ops
- Analytics & performance reporting

## 8 KEY PARTNERS

- Ministry of Agriculture & municipalities
- Pesticide suppliers & agrochemical
- Universities & research institutions
- IoT connectivity & cloud partners
- Distributors & agents in the Gulf

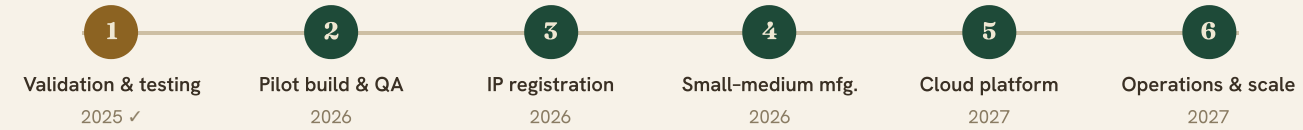
## 9 COST STRUCTURE

- R&D & product development
- Hardware components & manufacturing
- Pesticide cartridges & consumables
- Field operations & maintenance
- Marketing, sales & partnerships

## UNIT ECONOMICS

Prototype BOM ~26 JOD today (small-batch, off-the-shelf); target ~**10 USD/unit** at scale via bulk parts, PCB integration and simpler assembly — against **200–350 USD** of value per tree saved and prevention of **70%+** tree loss.

# From validated prototype to operational scale.



**Stage one is complete** — firmware, on-device AI, dashboard and the v4 enclosure and injection system are built and award-validated. Near-term priorities are to expand the real-world weevil audio dataset toward the **>92% target** after real-field calibration, run a Jordan Valley field pilot, ruggedise the enclosure, validate treatment dosing and obtain regulatory approval, and scale the mesh while driving unit cost down. The path runs in three stages: a competition prototype that safely demonstrates the full loop with clear water; a field pilot to expand the dataset and validate the live-infestation response; and regulated deployment with approved treatment, maintenance teams and ministry or farm partnerships.

FUNDING & NEXT MILESTONE

The team is raising a **15,000 JOD pre-seed** for pilot manufacturing, IP registration and testing, on a path to seed readiness and aligned with the global coalition against the Red Palm Weevil. The next technical milestone is **live-infestation validation**: testing the full detect-decide-act loop on confirmed infested palms under controlled agricultural supervision.

REFERENCES

[1] FAO. **The Red Palm Weevil threatens food security and livelihoods**. FAO Newsroom, 2019.

[2] FAO. **Current Situation of RPW in the NENA Region** — Scientific Consultation, Rome, 2017.

[3] CABI Plant Health Cases. **Management of RPW in Al-Ahsa Oasis**, 2023.

[4] The Jordan Times. **Campaign launched to eliminate palm tree pest**, 2025.

[5] Soroker, V. et al. Phytoparasitica. **Current status of RPW infestation in date palm**.

[6] Mashal, M. M. & Obeidat, B. F. (2019). **Efficacy of emamectin benzoate micro-injection to control RPW**. Heliyon 5(6), e01833. NARC, Jordan.

[7] **Early detection of RPW using a distributed optical sensor**. Scientific Reports, 2020.

[8] **Feeding or starvation: RPW susceptibility to insecticides**. Frontiers in Insect Science, 2025.

[9] Earth Island Journal. **How AI Could Save Date Palms**, 2024.

[10] **Early Detection of RPW using Deep Learning of Acoustic Signals**. arXiv:2308.15829, 2023.

[11] Koubaa, A. et al. (2020). **Smart Palm: An IoT Framework for RPW Early Detection**. Agronomy 10(7).

[12] Wang, B. et al. (2021). **Detecting RPW Using ML and Fiber-Optic Distributed Acoustic Sensing**. Sensors 21(5).

[13] EcoMENA. **Red Palm Weevil: A Real Threat to Date Palms in the Middle East**, 2025.